

Research Paper Writing and Research Tools

From Research Idea to Journal Publication

A 2-Day Hands-On Workshop

For PhD scholars · Research scholars · PG students · Early-career researchers

DAY 1

Research Fundamentals · Literature Search · Literature Review · Research Gap



How to Use This Deck

FOR THE FACILITATOR

- Every section follows the same rhythm:
Concept → Explanation → Example → Tool → Live Demo → Exercise → Common Mistakes → Takeaway
- **DEMO** slides = switch to a live browser/software window. A demo script is in the speaker notes.
- **ACTIVITY** slides = stop presenting, start the timer, walk the room.
- Speaker notes contain what to *say*, what to *show*, and what to *watch out for*.

FOR THE PARTICIPANT

- Handouts referenced on slides are in the `handouts/` folder:
 - `paper-extraction-template.md`
 - `literature-matrix-template.csv`
 - `research-gap-worksheet.md`
 - `journal-selection-checklist.md`
 - `reviewer-response-template.md`
 - `mini-proposal-template.md`
- You are expected to produce **your own artefacts** on your own topic — not to copy the worked examples.

Worked examples in this deck are illustrative composites built to teach structure. Bibliographic details of well-known public works (e.g. ResNet, BERT, U-Net, ViT) are used as examples of *type*; always verify any citation against the publisher record before using it in your own manuscript.



Two-Day Agenda

	DAY 1 — Idea to Research Gap	DAY 2 — Gap to Published Paper
09:30–11:00	S1 Research & publication fundamentals S2 Selecting a research topic	S9 Anatomy of a paper S10 Title · S11 Abstract
11:15–13:00	S3 Searching for papers (demo) S4 Reading papers efficiently	S12 Introduction S13 Related work
14:00–15:30	S5 Literature review S6 Literature matrix (demo)	S14 Methodology S15 Experimental design · S16 Metrics
15:45–17:30	S7 Finding the research gap S8 Objectives & contributions Day 1 activity + presentations	S17 Results/Discussion · S18 Figures/Tables S19 References · S20 AI tools · S21 Ethics S22 Journals · S23 Submission · S24 Reviewers S25 Case study + final activity

Continuous thread: you pick *one* topic on Day 1 morning and carry it through all 25 sections. By 17:30 on Day 2 you have a mini research proposal ready to show your supervisor.

Learning Outcomes

By the end of Day 2, a participant will be able to:

1. Distinguish a **research problem** from an engineering/project problem and state one in publishable form.
2. Build a **reproducible search string** and run it across Scopus, Web of Science, IEEE Xplore, ACM DL, Scholar and Semantic Scholar.
3. Read a paper in **three passes** and extract 14 structured fields.
4. Build and interpret a **literature matrix** over 10–15 papers.
5. Identify and *justify* a **research gap** using a named gap taxonomy.
6. Convert a gap into **RQs → objectives → hypotheses → contributions**.
7. Write each section of an **IEEE-style manuscript** to a checklist.
8. Design an experiment with **baselines, ablations and reproducibility controls**.
9. Choose **evaluation metrics** that match the problem, not the convenience.
10. Manage references in **Zotero/BibTeX/Overleaf** with zero citation errors.
11. Use **AI tools productively and defensibly**, with an explicit verification protocol.
12. Select a **legitimate target journal** and respond to reviewers professionally.



Section 1

Introduction to Research and Publication

What research is, what it is not, and the full path from idea to publication

What Is Research?

Research is a systematic, documented, and critically evaluated inquiry that produces knowledge which did not exist in the accessible record before, and which others can verify.

FOUR NON-NEGOTIABLE PROPERTIES

Property	Test question
Systematic	Could someone repeat your procedure from your description alone?
Novel	What is now known that was not known (or not <i>shown</i>) before?
Evidenced	What evidence supports each claim, and how strong is it?
Communicated	Is it in the permanent, citable, peer-reviewed record?

THE THREE QUESTIONS EVERY REVIEWER ASKS

- 1. What is new? (novelty)
- 2. Why should I care? (significance)
- 3. Why should I believe you? (validity)

If you cannot answer all three in **two sentences each**, you do not yet have a paper — you have an activity.

Research vs Project Development

Dimension	Project / Development	Research
Goal	Deliver a working artefact	Produce verifiable new knowledge
Success criterion	It works, on time, to spec	Claim is novel, significant, and validated
Question form	"How do I build X?"	"Under what conditions does X hold, and why?"
Comparison	Against requirements	Against baselines and state of the art
Evaluation	Testing / user acceptance	Controlled experiments, statistics, ablations
Failure	Bug to be fixed	Finding to be reported and explained
Reuse of others' work	Encouraged and sufficient	Necessary but insufficient — must be <i>exceeded</i>
Output	Software, report, demo	Paper, thesis, dataset, theorem, protocol

Overlap is normal. Most CS research *contains* a project. The project is the instrument; the knowledge claim is the product. A paper that reports only the instrument is a technical report, not a research paper.



Research Problem vs Project Problem

❌ PROJECT PROBLEM

"Our college needs a web system to detect plagiarism in student assignments."

- Bounded by *local* need
- Solved by integrating existing tools
- Success = deployment
- Generalisation irrelevant
- No baseline comparison required

❌ PROJECT PROBLEM DRESSED AS RESEARCH

"Design and implementation of a CNN-based plant disease detection system using transfer learning."

Why it fails: every verb is engineering. Nothing is unknown.

✅ RESEARCH PROBLEM

"Transformer-based plagiarism detectors are reported at >0.90 F1 on English benchmarks, but their behaviour on **code-mixed Hindi-English** student text is unmeasured; character-level obfuscation is hypothesised to degrade them disproportionately."

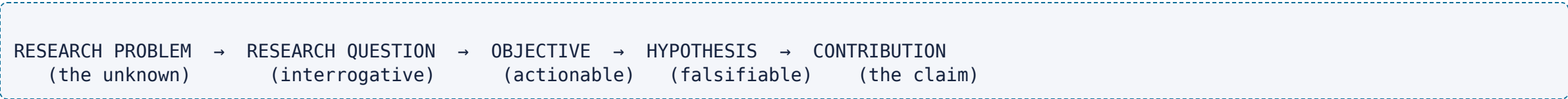
- Names a **specific unknown**
- Implies a **measurable** study
- Implies **baselines** (existing detectors)
- Result is informative **whichever way it goes**

THE 4-PART TEST FOR A RESEARCH PROBLEM

1. Is the answer **currently unknown** in the literature?
2. Can it be **measured or proven**?
3. Would a **negative result still be publishable**?
4. Does it matter to someone **outside your institution**?



From Problem to Hypothesis: The Five Statements



Statement	Function	Grammatical form	Worked example (code-mixed plagiarism detection)
Problem	Names the unknown + why it matters	Declarative, with a gap marker	"Detector robustness on code-mixed text is unquantified, so institutions in multilingual settings cannot trust off-the-shelf tools."
Question	Makes the unknown answerable	Interrogative, one variable pair	" RQ1 : How much does F1 of SBERT-based detection drop from monolingual to code-mixed paraphrase, and does script normalisation recover it?"
Objective	Commits to work	"To + verb + object + condition"	"To quantify F1 degradation of three detectors on a 4,000-pair code-mixed benchmark under four obfuscation levels."
Hypothesis	Predicts, and can be wrong	H0 / H1 with a measurable effect	" H1 : Transliteration-normalised embeddings improve F1 by ≥ 0.05 over raw code-mixed input (paired, $\alpha=0.05$)."
Contribution	What the field gains	Noun phrase, calibrated	"A public code-mixed paraphrase benchmark and the first quantification of script-induced degradation in SBERT detectors."

Methodology and Contribution

Research methodology

Not "the tools I used". It is the **defensible logic** connecting your question to your evidence.

Layer	Question it answers
Design	Comparative? Ablative? Observational? Proof-based?
Data	Which datasets, why those, what splits, what biases
Procedure	Preprocessing, model, training, tuning protocol
Controls	Baselines, seeds, fixed budgets, blinding
Analysis	Metrics, statistical tests, error analysis
Threats	Validity limits and how you mitigate them

"Methodology = Python + TensorFlow + Colab" is an *environment*, not a methodology.

Research contribution

The **transferable** thing the field keeps after your paper.

Contribution type	Example
New method	A loss function that handles label noise
New theory/analysis	Convergence bound for that loss
New empirical knowledge	Site-wise splits cut reported AUC by 0.18 across 6 published models
New resource	An annotated code-mixed benchmark
New evaluation	A protocol that detects shortcut learning
Synthesis	A systematic review that reconciles contradictory findings
Reproduction/refutation	Failure to reproduce a headline result, with diagnosis

Types of Research

Experimental vs theoretical

Experimental	Manipulate variables, measure outcomes. <i>Most ML papers.</i>
Theoretical	Derive results from assumptions; proofs, bounds, complexity.
Computational/simulation	Study systems too costly to build physically.
Observational/empirical	Measure what exists without intervening (mining 10k GitHub repos).

Applied vs fundamental

Fundamental (basic)	Understand a phenomenon. <i>Why do transformers generalise despite overparameterisation?</i>
Applied	Solve a specified practical problem. <i>Detect diabetic retinopathy on low-cost fundus cameras.</i>
Translational	Bridge the two; deployment constraints become research variables.

Quantitative vs qualitative

Quantitative	Numbers, statistics, generalisation. <i>Accuracy across 5 seeds.</i>
Qualitative	Meaning, mechanism, experience. <i>Interviews with 12 radiologists on trust in AI output.</i>
Mixed methods	Both, deliberately integrated. Strong for HCI/SE/health-AI.

Why this classification matters practically: it determines your **validity threats**, your **sample-size logic**, your **reporting standards** (e.g. CONSORT-AI for clinical trials, PRISMA for systematic reviews, ACM/IEEE artefact badging for systems), and often your **target venue**.

What Is a Research Paper?

A research paper is an argument, supported by evidence, formatted as a permanent citable record.

THE ARGUMENT SKELETON HIDING IN EVERY PAPER

Something matters	→ Introduction ¶1
It is not solved / not known	→ Introduction ¶2-3
Prior work falls short because	→ Related work
So we did this, carefully	→ Methodology
Here is what we observed	→ Results
Here is what it means, and its limits	→ Discussion
Therefore this is now known	→ Conclusion

Every section exists to defend one link in that chain. If a paragraph defends no link, delete it.

NOT RESEARCH PAPERS

- Technical report / white paper
- Project report / thesis chapter (until reworked)
- Tutorial or blog post
- Extended abstract or poster
- Preprint (a *stage*, not peer review)
- Patent (protection, not knowledge claim)

A preprint on arXiv establishes **priority and visibility**, not peer-reviewed status. Most journals allow it; **always check the specific journal's preprint policy** before posting.



Conference Paper vs Journal Paper

Dimension	Conference paper	Journal paper
Review	Single round, fixed deadline, PC-based	Multiple rounds, rolling, editor + referees
Decision	Accept / reject (sometimes minor revision)	Major/minor revision cycles; can iterate for months
Turnaround	2–4 months to decision, fixed presentation date	3–12+ months to first decision
Length	6–10 pages, hard page limit	8–35 pages, often flexible + appendices
Depth expected	A crisp new idea, adequate validation	Complete study: exhaustive baselines, ablations, statistics, related work
Prestige in CS/AI	Very high — NeurIPS/ICML/ICLR/CVPR/ACL/AAAI often outrank journals	High — required in most non-CS fields and by many PhD regulations
Extension	Often extended into a journal version (typically ≥30% new material ; check policy)	Usually terminal
Cost/effort	Registration + travel; camera-ready fast	APC may apply; sustained revision effort

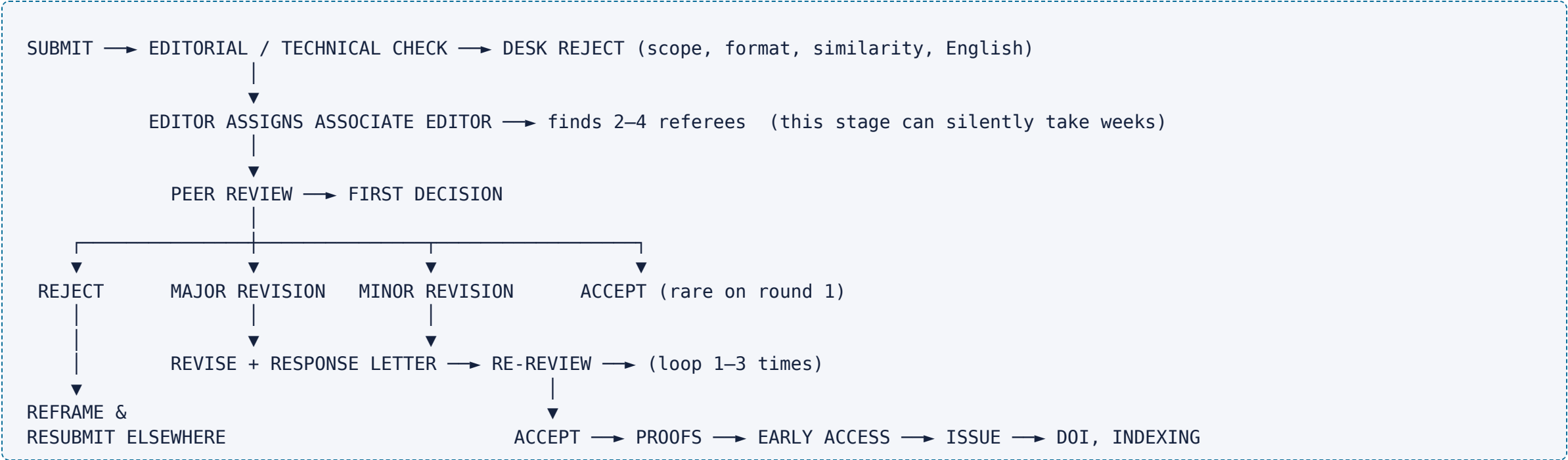
Check your university's PhD regulations first. Many Indian and European universities require *N* SCI/SCIE- or Scopus-indexed **journal** papers regardless of how strong your conference record is. Plan the mix deliberately.

Research Paper vs Review vs Survey

	Primary research paper	Review paper	Survey / Systematic review
Unit of study	Data, systems, subjects	Published literature	Published literature, with a protocol
New experiments?	Yes	Usually no	No (but may include meta-analysis)
Contribution	New result	Interpretation, taxonomy, agenda	Reproducible synthesis of evidence
Method section	Datasets, models, protocol	Selection rationale	Explicit protocol: databases, strings, dates, inclusion/exclusion, PRISMA flow, quality appraisal
Typical length	8–15 pages	15–30	20–60
Who should write it	Anyone with a result	Usually experienced researchers, often invited	A team; suitable for a first PhD paper if done rigorously
Risk	Reviewer disputes validity	"Just a list of papers" → reject	Protocol not reproducible → reject

A rigorous systematic review is an excellent first publication: it forces mastery of the literature, produces your literature matrix *and* your research gap as by-products, and is citable for years. Follow **PRISMA 2020** (Page et al., *BMJ*, 2021) or **Kitchenham's guidelines** for software engineering.

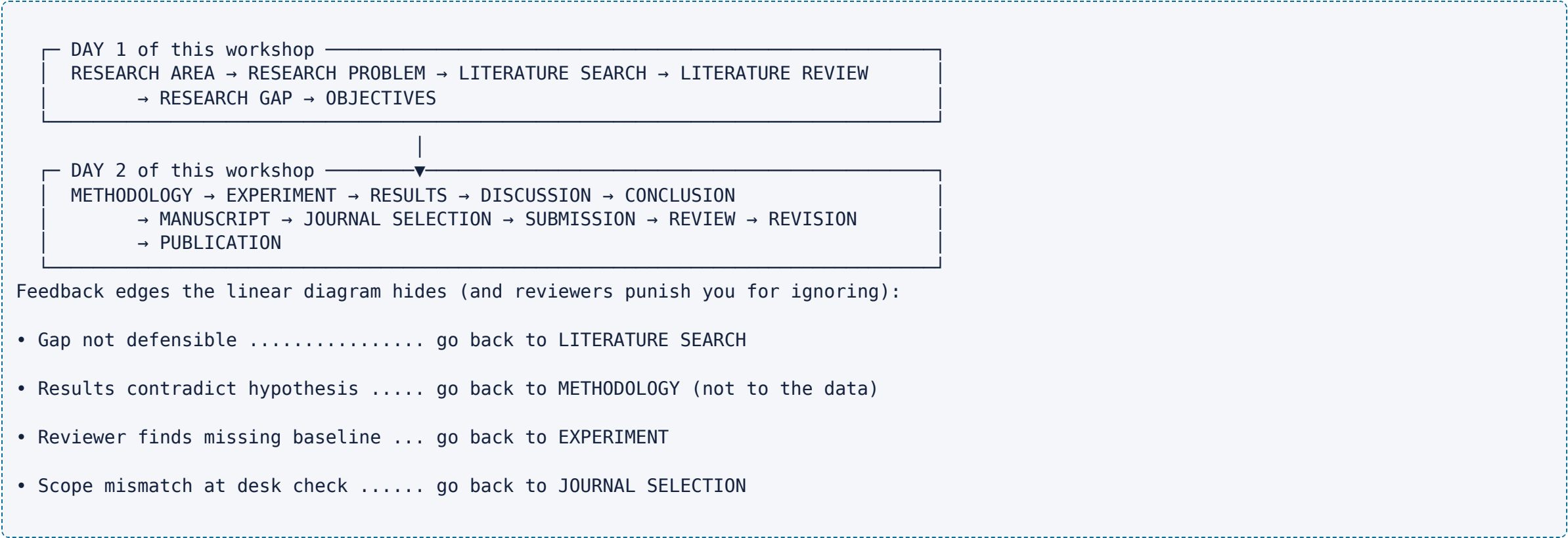
The Journal Publication Lifecycle



Stage	Typical duration	What you control
Desk check	3 days – 3 weeks	Scope fit, formatting, similarity report, language quality
Under review	1 – 6 months	Nothing — but you may send a polite status query after the journal's stated period
Revision	You are given 30–90 days	Response quality; ask for an extension <i>before</i> the deadline
Production	2 – 8 weeks	Proof accuracy (this is your last chance to fix errors)



The Complete Research Workflow



Phase	Realistic time for a first paper	Where beginners under-invest
Area → gap (Day 1 skills)	4–10 weeks	Almost always rushed — the root cause of later rejection
Method → experiments	8–20 weeks	Baselines and ablations skipped
Writing → submission	3–6 weeks	Abstract, related work, response letter
Review → publication	3–12 months	Emotional preparation



Section 1 — Wrap-Up

COMMON MISTAKES

- Calling an implementation a contribution
- Writing objectives as activity titles ("Study of...")
- Choosing a venue after writing, not before
- Treating the workflow as strictly linear
- Believing a preprint is peer reviewed
- Extending a conference paper without disclosing it

RECOMMENDED TOOLS FOR THIS STAGE

Tool	Purpose
Zotero / Mendeley	Reference library from day one
Notion / Obsidian / OneNote	Research journal, decision log
Overleaf	LaTeX manuscript + version history
Git + GitHub	Code, data scripts, artefact release

ACTIVITY 1.1 — 10 MIN

"Project or research?"

1. Write your current work in **one sentence**.
2. Classify it: project / research / unclear.
3. If project or unclear, write the knowledge claim it could support:
"We do not currently know whether _____ , and this study would show it."
4. Swap with a neighbour. The neighbour applies the 4-part test and marks each item pass/fail.

Takeaway: A paper is not a record of what you built. It is a defended claim about what is now known — and the claim must be chosen before the building starts.

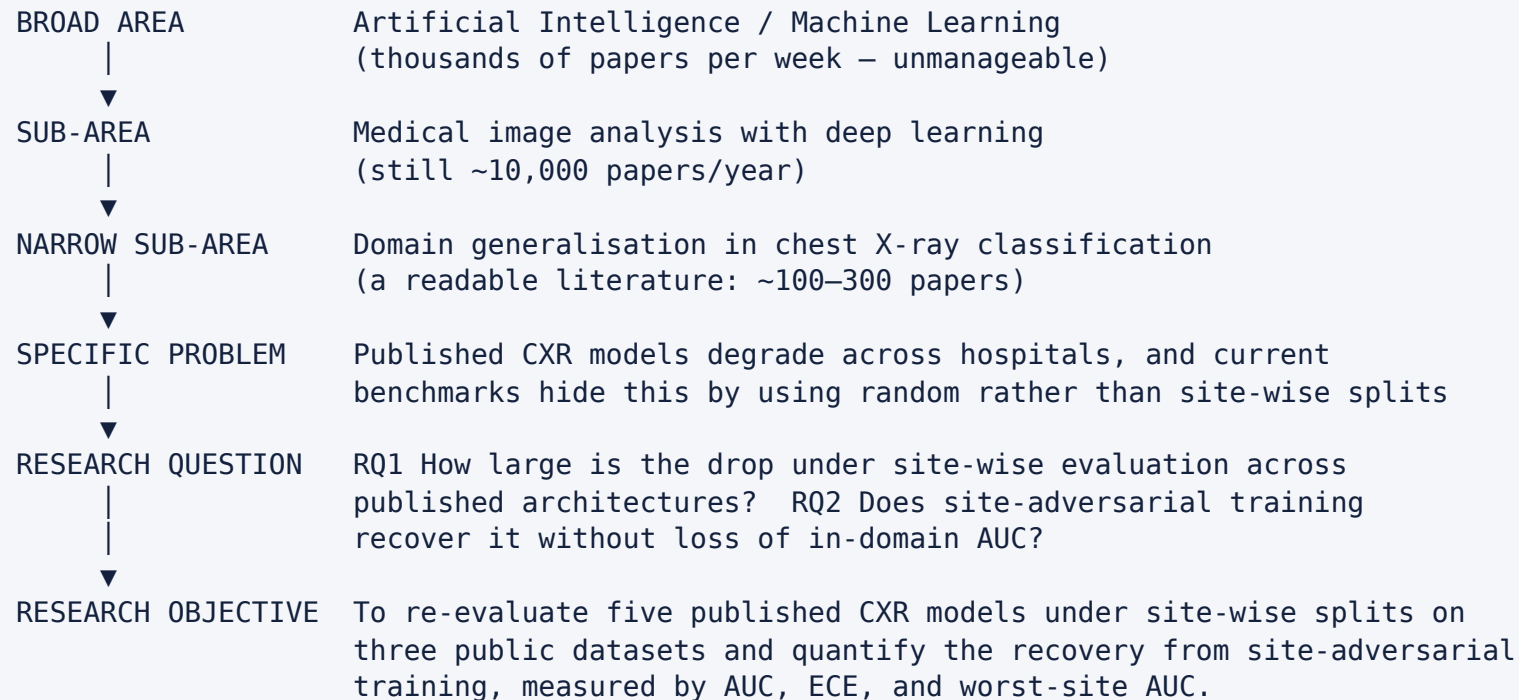


Section 2

How to Select a Research Topic

From a broad area to a problem you can finish, defend, and publish

The Narrowing Funnel



Test of adequate narrowing: you can name the **datasets**, the **baselines**, and the **metrics** in one breath. If you cannot, you are still one level too broad.

How to Select a Research Area

Choose at the intersection of **five** constraints — not on interest alone.

Constraint	Question to answer honestly	Failure if ignored
Personal endurance	Can I read this literature for 3–5 years without resentment?	Abandonment at year 2
Supervisor capability	Can my supervisor actually critique my technical work here?	Unsupervised drift, weak papers
Local resources	Do I have the data, GPUs, subjects, licences, ethics approval?	Stalled experiments
Field momentum	Is publication volume rising, flat, or collapsing?	Working in a dead area, or in a saturated hype peak
Career market	Do jobs/funding calls name this area?	Employability mismatch

MOMENTUM CHECK (10 MINUTES, DO IT NOW)

1. Scopus → search area keywords → **Analyze search results** → *Documents by year*.
2. Rising and not yet vertical = good entry point.
3. Vertical (e.g. "LLM agents" 2023–2026) = high visibility, brutal competition, fast obsolescence.
4. Flat/declining = easier novelty, fewer readers, harder to publish in high-IF venues.

WHERE COMPETITIVE NOVELTY ACTUALLY LIVES

- **Intersections**: your area × another field's constraint (privacy, low-resource languages, edge compute, clinical workflow, agriculture, regional data)
- **Neglected populations/domains** in an otherwise mature area
- **Evaluation and reproducibility** of hyped methods
- **Efficiency** frontiers of large models
- Anywhere a paper says *"we leave this to future work"*



Identifying Emerging Topics and Unresolved Problems

EMERGING-TOPIC RADAR **TOOLS**

Source	What to look at	Cadence
arXiv (cs.LG, cs.CV, cs.CL)	New listings; set Google Alerts on key phrases	Weekly
Semantic Scholar	Feeds, "Highly Influential Citations", citation velocity	Weekly
Connected Papers	Graph around a seed paper: <i>Prior / Derivative</i> works	Per topic
Litmaps / ResearchRabbit	Auto-updating literature maps with alerts	Monthly
Scopus / WoS	Documents-by-year trend; top source titles; author networks	Per topic
Journal CFPs	Special issue calls = editors publicly declaring gaps	Monthly
Workshop titles at NeurIPS/CVPR/ACL	The field's 12-month agenda	Yearly
Benchmark leaderboards (Papers with Code, OpenML)	Where progress has <i>stalled</i>	Per topic

SIX PLACES UNRESOLVED PROBLEMS ARE STATED *EXPLICITLY*

1. **"Limitations"** section of recent papers
2. **"Future work"** in conclusions
3. **Reviewer-visible weaknesses** on OpenReview (ICLR/NeurIPS) — free access to expert criticism
4. **Rebuttals**: what authors admitted they could not do
5. **Survey papers**: their "open challenges" section
6. **Reproducibility reports** (ML Reproducibility Challenge, ReScience)

OpenReview is the most underused resource in this list.

Read the reviews of accepted *and rejected* papers in your area: you learn the field's actual standards of evidence, in the field's own words.



Mining Limitations From Papers Into Topics

PROCEDURE (REPEAT OVER 10–15 RECENT PAPERS)

- 1. Open the paper's **Limitations**, **Threats to validity**, **Discussion** and **Conclusion/Future work**.
- 2. Copy each admitted weakness into a spreadsheet **verbatim**, with paper + section + page.
- 3. Tag it: **data** / **method** / **evaluation** / **scale** / **generality** / **explainability** / **efficiency** / **ethics**.
- 4. **Count tags across papers**. A weakness admitted by 6 of 15 papers is a *field-level* gap, not one author's excuse.
- 5. For each recurring weakness ask: *is it unsolved, or solved elsewhere and simply not applied here?*
 - Unsolved → potential **novel contribution**
 - Solved elsewhere → potential **transfer/application contribution** (still publishable; be honest about the framing)

WORKED MINING EXAMPLE

"Our evaluation is limited to English; performance on morphologically rich languages remains unexplored." — *typical NLP paper*

Reading	Resulting topic
Naïve	"Apply the method to Hindi" (weak: pure application)
Sharp	" Which property of the method breaks on morphologically rich languages — the tokeniser, the positional scheme, or the pretraining corpus? Isolate it by ablation across 4 language families."

Do not stop at "*apply X to Y*". Ask **why** X would fail on Y, then measure that mechanism. Mechanism turns application into science.

Avoiding an Overly Broad Topic

❌ Too broad	Why it fails	✅ Narrowed and defensible
"Deep learning for healthcare"	No question, no boundary, no finishable study	"Site-wise generalisation of CXR classifiers across three public datasets"
"Improving NLP for Indian languages"	22 official languages, dozens of tasks	"Tokeniser-induced degradation in Marathi and Kannada extractive QA, with a morphology-aware BPE variant"
"AI in agriculture"	Domain, not problem	"Few-shot leaf-disease detection under field lighting with ≤ 50 labelled images per class on a Jetson-class device"
"Blockchain for security"	Two buzzwords, no unknown	"Gas-cost and latency limits of on-chain revocation for verifiable academic credentials at 10^5 users"
"Explainable AI"	Whole research field	"Do Grad-CAM saliency maps agree with radiologist-annotated regions in CXR pneumonia cases, and does agreement predict model correctness?"

NARROWING OPERATORS — APPLY UNTIL IT FITS

Population (which subjects/domains) · **Task** (which exact output) · **Condition** (noise, low-light, low-resource, adversarial) · **Constraint** (latency, memory, privacy, labels) · **Comparison** (against which baselines) · **Outcome** (which metrics)

THE ONE-BREATH TEST

"I compare *[methods]* on *[datasets]* under *[condition]* measured by *[metrics]*, to find out whether *[question]*."

If you cannot fill all five slots, you are not ready to start experiments.

Is the Problem Research-Worthy?

THE 7-QUESTION GATE

#	Question	Fail signal
1	Is the answer unknown in the literature?	You find 5 papers already answering it
2	Is it measurable/provable?	No metric, no proof strategy
3	Is a negative result publishable?	Only "we win" is reportable
4	Can it be done with my resources in my time?	Needs 512 GPUs or 5,000 patients
5	Will anyone cite it?	No community, no benchmark, no application
6	Is it ethically and legally doable?	No consent, no licence, no IRB path
7	Can I state the contribution in one sentence?	Vague, plural, or grandiose

Novelty vs improvement

	Novelty	Improvement
Claim	"This is a different way of thinking about the problem"	"This is measurably better at the same task"
Evidence	Conceptual argument + empirical support	Rigorous comparison + statistics + ablation
Reviewer risk	"Is it <i>sound</i> ?"	"Is it <i>significant</i> ?" — the 0.3% problem

Both are publishable. Improvement papers succeed when the gain is (a) statistically significant across seeds, (b) explained mechanistically by an ablation, and (c) not bought with hidden extra compute/data/tuning.

+0.4% accuracy on one dataset, one seed, no significance test, against a weakly tuned baseline = **the most common rejection in ML.**



Feasibility Audit — Do This Before You Commit

Dimension	What to verify now	Red flag	Where to check
Dataset exists	Named dataset, size, labels, licence, access path	"We will collect our own data" with no plan	Kaggle, HuggingFace Datasets, UCI, PhysioNet, OpenML, Papers with Code, Zenodo, government portals
Licence permits use	Redistribution and commercial terms; CC-BY vs CC-BY-NC vs custom DUA	Dataset is behind a DUA needing a PI signature	Dataset landing page / licence file
Ethics/consent	IRB/IEC approval need, anonymisation, GDPR/DPDP constraints	Human/patient data, no approval route	Institutional ethics committee
Compute budget	GPU-hours for your model × baselines × ablations × seeds	"Fine-tune a 70B model" on one 8 GB GPU	Your lab; Colab/Kaggle free tiers; institutional HPC
Baselines available	Official code + weights; do they run?	No code and no reimplementation in 2 weeks	GitHub, Papers with Code, HuggingFace
Evaluation defined	Metrics, splits, statistical test, significance plan	"We'll see the accuracy"	Benchmark papers, S15–S16 of this workshop
Time	Working backwards from your submission deadline	No slack for failures (assume 40%)	Your Gantt chart

Compute arithmetic, done honestly: 4 baselines + your model = 5 configs × 5 seeds × 3 datasets = **75 training runs**. At 2 GPU-hours each = 150 GPU-hours, *plus* tuning, *plus* the ablation study (often 2–3× more), *plus* reruns after reviewers ask. Budget 3× your first estimate.



Section 2 — Wrap-Up

COMMON MISTAKES

- Choosing a topic from a title, not from reading
- Confusing a *domain* with a *problem*
- Ignoring dataset licences and ethics until submission
- Ignoring baseline availability until the last month
- Chasing hype without a durable core contribution
- "Apply X to Y" with no mechanism question
- No feasibility arithmetic for compute or time

TOOLS RECAP TOOLS

Scopus/WoS trends · arXiv + Google Alerts · Semantic Scholar feeds · Connected Papers / Litmaps / ResearchRabbit · OpenReview · Papers with Code · HuggingFace Datasets · journal special-issue CFPs

DELIVERABLE FROM SECTION 2 (KEEP IT, YOU WILL USE IT ALL WORKSHOP)

Area: _____

Sub-area: _____

Problem: _____

RQ1/RQ2: _____

Objective: _____

Datasets: _____ (licence: _____)

Baselines: _____ (repo: _____)

Metrics: _____

Compute est: _____ GPU-hours

Killer risk: _____

Takeaway: A good topic is *narrow enough to finish, general enough to matter, and specific enough to name its datasets, baselines and metrics in one sentence.*

Section 3

How to Search for Research Papers

Reproducible searching: strings you can put in a paper, not clicks you cannot remember

The Database Landscape

Platform	What it indexes	Access	Strength	Limitation
Google Scholar	Everything: journals, conferences, preprints, theses, patents	Free	Widest recall; full-text search; "Cited by"	No quality control; no export of result sets; weak field syntax; irreproducible result counts
Scopus (Elsevier)	~28k+ peer-reviewed sources, curated	Subscription	Best query syntax + analytics; author/affiliation profiles; CiteScore	Coverage gaps for CS conferences and books
Web of Science (Clarivate)	Core Collection: SCIE, SSCI, A&HCI, CPCI, ESCI	Subscription	Highest curation; JCR Impact Factor ; citation-report analytics	Narrower than Scopus; complex editions
IEEE Xplore	IEEE + IET journals, conferences, standards	Subscription	Essential for EE/CS; standards; Command Search	Publisher-limited
ACM Digital Library	ACM journals, conferences, SIGs	Subscription	Definitive for core CS venues	Publisher-limited
ScienceDirect	Elsevier full text	Subscription	Full-text of Elsevier journals	One publisher
SpringerLink	Springer/Nature journals, LNCS, books	Subscription	LNCS conference proceedings; books	Weak advanced syntax (no truncation)
Semantic Scholar	~200M+ records, AI-enhanced	Free	TLDRs, citation <i>intent</i> , influential citations, open API, alerts	Metadata noise
PubMed / PMC	Biomedical	Free	MeSH controlled vocabulary	Health only
OpenAlex / Lens.org / Dimensions / CORE / BASE	Open bibliographic graphs	Free tiers	Excellent when you have no subscription	Less curation than Scopus/WoS

Never search only one platform. A systematic search uses ≥ 3 databases plus one grey-literature source (arXiv), and reports each string and each date.



Step 1: Turn a Question Into Concepts and Synonyms

Question: *Do deep models for chest X-ray classification generalise across hospitals, and do domain-generalisation methods help?*

Concept block	Preferred terms	Synonyms / variants / spellings	Narrower terms
C1 Population/data	chest radiograph	"chest X-ray", CXR, "thoracic radiograph", "chest radiography"	CheXpert, MIMIC-CXR, NIH ChestX-ray14, PadChest
C2 Task	classification	detection, diagnosis, screening, "multi-label classification"	pneumonia, pneumothorax, cardiomegaly
C3 Method	"deep learning"	CNN, "convolutional neural network", transformer, "vision transformer", ViT, DenseNet	ResNet, EfficientNet, Swin
C4 Phenomenon (the core!)	"domain generalisation"	"domain generalization", "domain shift", "distribution shift", "out-of-distribution", OOD, "external validation", "cross-institution", "site effect", "dataset bias"	"site-wise split", "hospital-level split"
C5 Outcome	AUC	AUROC, "area under the curve", sensitivity, specificity, calibration	"worst-group accuracy", ECE

RULES

- One block per **concept**, joined by **OR** inside, **AND** between.
- Include **both** -ise/-ize spellings and British/American variants.
- Include **acronyms and their expansions**.
- Include **dataset names** — they retrieve papers that never use your abstract vocabulary.

WHERE SYNONYMS COME FROM **TOOLS**

- Keywords of 5 papers you already trust
- **Index terms** on IEEE Xplore; **Author Keywords** in Scopus
- **MeSH** (PubMed) / **ACM CCS** concepts / **IEEE Thesaurus**
- Titles of the papers your seed paper cites
- Ask an LLM for synonym candidates — then **verify each one retrieves real, relevant papers** before trusting it

Step 2: Boolean Operators and Field Codes

UNIVERSAL LOGIC

Operator	Effect	Note
AND	Both — narrows	Default on most platforms
OR	Either — widens	Wrap OR blocks in parentheses
AND NOT / NOT	Excludes — use sparingly	Silently kills relevant papers
" "	Exact/loose phrase	Essential for multiword concepts
()	Grouping	Precedence differs per platform — always parenthesise
*	Truncation: <code>general^{li}*</code> → generalise/generalize/generalisation	Not on SpringerLink; Scholar unreliable
?	Single character: <code>wom?n</code>	Scopus, WoS, IEEE
W/n , NEAR/n , ONEAR/n	Proximity within <i>n</i> words	Scopus W/n , WoS NEAR/n , IEEE NEAR/n (ONEAR/n = ordered)
PRE/n	First term precedes second by $\leq n$	Scopus, ScienceDirect

FIELD CODES WORTH MEMORISING

Platform	Field syntax
Scopus	TITLE-ABS-KEY(), TITLE(), AUTHKEY(), SRCTITLE(), DOCTYPE(ar), PUBYEAR > 2021
Web of Science	TS= (topic), TI= , AB= , AK= , SO= , PY=2022-2026 , DT= (Article)
IEEE Xplore (Command Search)	("Document Title":...), ("Abstract":...), ("Index Terms":...), ("All Metadata":...)
ScienceDirect	tak(...) title-abs-key; ≤ 8 Boolean operators per field
Google Scholar	intitle: , author: , source: , -term ; ~256-character limit

Platform syntax changes. **Verify against the platform's own help page on the day you search**, and paste the exact string + date into your notes.



Step 3: Build the Search String DEMO

SCOPUS (THE REFERENCE IMPLEMENTATION — BUILD HERE, THEN PORT)

```
TITLE-ABS-KEY(  
  ("chest X-ray*" OR "chest radiograph*" OR CXR OR "thoracic radiograph*" OR CheXpert OR "MIMIC-CXR" OR "ChestX-ray14" OR PadChest)  
  AND ("deep learning" OR CNN OR "convolutional neural network*" OR transformer* OR "vision transformer" OR ViT)  
  AND ("domain generalization" OR "domain shift" OR "distribution shift" OR "out-of-distribution" OR OOD OR "external validation" OR "cross-institution*" OR "site effect*" OR "dataset bias")  
)  
AND PUBYEAR > 2021 AND PUBYEAR < 2027  
AND (LIMIT-TO(DOCTYPE,"ar") OR LIMIT-TO(DOCTYPE,"cp"))  
AND (LIMIT-TO(LANGUAGE,"English"))
```

PORTED TO WEB OF SCIENCE

```
TS=((("chest X-ray*" OR "chest radiograph*" OR CXR  
  OR CheXpert OR "MIMIC-CXR")  
  AND ("deep learning" OR CNN OR transformer*)  
  AND ("domain generalization" OR "domain shift"  
    OR "distribution shift" OR "out-of-distribution"  
    OR "external validation"))  
AND PY=2022-2026
```

PORTED TO IEEE XPLORE COMMAND SEARCH

```
("All Metadata":"chest X-ray" OR "All Metadata":CXR  
  OR "All Metadata":CheXpert)  
AND ("All Metadata":"deep learning"  
  OR "All Metadata":transformer)  
AND ("All Metadata":"domain shift"  
  OR "All Metadata":"domain generalization"  
  OR "All Metadata":"out-of-distribution")
```

Then set **Year: 2022–2026**, **Content type: Journals / Conferences**.



Effective vs Ineffective Queries

✗ Ineffective	Why it fails	✓ Effective
deep learning chest x ray	No phrases, no synonyms, no field limit → 100k+ noisy hits	TITLE-ABS-KEY(("chest X-ray*" OR CXR) AND "deep learning")
how can I improve generalisation of CNN models across hospitals?	Natural-language question in a Boolean database	Concept blocks joined by AND/OR (previous slide)
"domain generalisation" only	One spelling → silently loses most of the literature	"domain generali*ation" OR "domain shift" OR "out-of-distribution"
CXR AND DL AND DG	Ambiguous acronyms (DL = deep learning? downlink? Dice loss?)	Acronym plus expansion, in an OR block
(A OR B) AND (C OR D) NOT COVID NOT survey NOT review	Over-exclusion; loses methodology papers and your own related-work sources	Include everything; screen by title/abstract instead
chest x-ray classification 2026	Year typed as a keyword, not a filter	Use PUBYEAR > 2021 / PY=2022-2026
One database only	Coverage bias; not reproducible	≥3 databases + arXiv, all strings logged
Sorting by relevance only	Buries recent work	Run both "sort by citations" and "sort by date"

Precision-recall trade-off, stated practically: aim for a set of **80–300 records** for a focused review. Above 500, add a concept block or tighten to titles. Below 40, drop a block, add synonyms, or widen the year window — and check your spelling variants first.

Filters, Citation Signals, and Progressive Refinement

FILTERS THAT MATTER

Filter	Use
Year	Methods: 2022–2026. Foundations: no limit (you <i>must</i> cite the origin)
Document type	Article (<i>ar</i>), Conference paper (<i>cp</i>), Review (<i>re</i>) — search reviews first , then primary studies
Subject area	Removes cross-domain false hits
Source title	Restrict to your target journals to learn their style and cite them
Open access	Only when you lack full-text access — never as a quality filter
Language	Declare it as a limitation if you restrict to English

READING CITATION COUNTS HONESTLY

- Counts are **age-confounded**: a 2025 paper with 8 citations may be hotter than a 2019 paper with 60.
- Use **citations/year**, and Semantic Scholar's *Highly Influential Citations* (citation **intent**, not just count).
- A highly cited paper may be highly *criticised*. Check **who cites it and why**.
- Self-citation clusters and citation cartels exist — inspect the citing set.

PROGRESSIVE REFINEMENT LOOP DEMO

- 1 Reviews/surveys 2022–2026 ↴
↓ harvest their taxonomies + terms
- 2 Rebuild string with better terms
↓
- 3 Sort by CITATIONS → find the 5–8 canonical papers
↓
- 4 Sort by DATE → find the 2025–2026 frontier
↓
- 5 For each key paper:
 - BACKWARD: mine its reference list
 - FORWARD : "Cited by" (Scholar/Scopus/WoS)
 - SIDEWAYS: Connected Papers graph↓
- 6 Stop when new searches return only papers you have already seen ← SATURATION
↓
- 7 Save the search + set an EMAIL ALERT

Saturation is the stopping rule. Not "when I have 15 papers" — when the literature stops surprising you.

Section 3 — Wrap-Up

COMMON MISTAKES

- Typing a sentence into a Boolean database
- One spelling, one acronym, one database
- Over-using NOT
- Year typed as a keyword
- Not logging the string and the search date
- Never running a backward + forward citation chase
- Never setting an alert
- Downloading 200 PDFs and reading none

SEARCH LOG TEMPLATE (PUT THIS IN YOUR PAPER)

Field	Value
Database	Scopus
Search date	12 March 2026
String	(verbatim)
Filters	2022–2026; ar, cp; English
Records	412
After screening	118 → 34 → 15

ACTIVITY 3.1 — 20 MIN

Build and run your own string

1. Write your concept/synonym table (≥ 3 blocks, ≥ 3 synonyms each, include dataset names). 5 min
2. Compose a Scopus- or WoS-style string. 5 min
3. Run it on ≥ 2 platforms; record record counts in the log. 5 min
4. Refine until you have 80–300 records; export RIS/BibTeX. 5 min
5. Shortlist 10–15 papers for tomorrow's matrix (mix: 2–3 reviews, 5–8 recent primary, 2–3 highly cited foundations).

Takeaway: A search is a *documented, reproducible instrument*. If you cannot paste your search string into your methods section and have someone reproduce your record count, you have browsed — you have not searched.



Section 4

How to Read a Research Paper Efficiently

Three passes, one template, zero wasted PDFs

The Three-Pass Method

Based on the widely used approach of [S. Keshav, "How to Read a Paper," ACM SIGCOMM CCR, 2007](#) — adapted here with an ML-specific extraction template.

	Pass 1 — Triage	Pass 2 — Comprehension	Pass 3 — Deep / reconstructive
Time	5–10 min	45–60 min	2–5 h (or days)
Read	Title, abstract, intro, section headings, all figures/tables + captions, conclusion, references skim	Everything except heavy proofs/derivations; all figures and tables carefully	Everything, including proofs, appendices, supplementary, and the code
Goal	Decide: discard / cite-only / read fully	Understand <i>what</i> they did and <i>whether the evidence supports it</i>	Understand <i>why</i> it works; be able to reimplement or refute it
Output	1 line in your triage sheet	Filled extraction template + matrix row	Reimplementation, notes on hidden assumptions, a new research idea
Applies to	Everything you download (100%)	Your core set (10–20%)	3–6 papers per project (2–5%)
Question	"Is this relevant to <i>my</i> question?"	"Do I believe the claim?"	"Can I build on, extend, or break this?"

Never read linearly from page 1 to page 12 on first contact. You will spend 40 minutes on a paper you should have discarded in 5 — and you will absorb the authors' framing before you have evaluated it.



Pass 1 — Triage in 5–10 Minutes

READ IN THIS ORDER

1. **Title + venue + year** → is it peer reviewed? which community?
2. **Abstract** → problem, method, headline result
3. **Figure 1 / architecture diagram** → the idea in one image
4. **All tables** → what was compared, on what data
5. **Section headings** → the shape of the argument
6. **Conclusion + limitations** → their own admission of weakness
7. **References**: do you recognise the canon? Is anything suspiciously absent?

THE FIVE TRIAGE ANSWERS TO WRITE DOWN

Problem · Method in ≤8 words · Datasets · Headline metric ·
Relevance to me: core / context / discard

DISCARD SIGNALS (BE RUTHLESS)

- Different task/domain with no transferable mechanism
- No baseline comparison at all
- No dataset named, or an unnamed private dataset with no access
- Venue is unindexed / suspected predatory (S22)
- Results impossible or unexplained (99.9% on a hard benchmark)
- Paper is superseded by a later, better version you already have

KEEP-BUT-ONLY-CITE SIGNALS

- Provides a definition, dataset, or statistic you need
- Is the canonical origin of a method you will use
- Contradicts another paper you are keeping — **flag it**, contradictions are gap material (S7)

TOOLS **Zotero** (save with PDF + metadata; colour tags: ■ core, ■ context, ■ discard) · **Semantic Scholar TLDR** for a one-line gist · **Connected Papers** to find the version you should have read instead · **Scholar "Cited by"** to check whether it was refuted.



Pass 2 — Comprehension in 45–60 Minutes

PROCEDURE

- 1. Re-read the abstract; write **their claim in your own words**. If you cannot, that is the paper's writing problem — note it, then dig.
- 2. Methodology: draw **their pipeline as a block diagram yourself**. Drawing exposes gaps their prose hides.
- 3. Experimental setup: list datasets, splits, baselines, metrics, hyperparameters, hardware, seeds.
- 4. Results: for **each table**, ask
 - What is the comparison?
 - Is the baseline **fairly tuned**?
 - Is the gain **larger than the variance**?
 - Are **all** datasets reported, or only the favourable ones?
- 5. Mark unknown terms/citations → build a **follow-up reading queue**.
- 6. Fill the extraction template (next slide).

CRITICAL-READING QUESTIONS THAT EXPOSE WEAK PAPERS

Target	Question
Novelty	Which <i>exact</i> component is new? Which existing paper is closest?
Fairness	Same data, splits, preprocessing, and tuning budget for all methods?
Leakage	Any patient/site/user appearing in both train and test? Any tuning on the test set?
Variance	Multiple seeds? Std devs? Confidence intervals? Any significance test?
Selectivity	Are the omitted datasets/metrics the inconvenient ones?
Cost	Parameters, FLOPs, latency, training time reported?
Reproducibility	Code and weights available? Do they match the paper?
Explanation	Do the authors explain <i>why</i> it works, or only <i>that</i> it wins?



Pass 3 — Deep / Reconstructive Reading

WHEN TO INVEST

- The paper is your **primary baseline**
- You intend to **extend** its method
- You suspect the result is **wrong** or **overstated**
- It is the theoretical foundation of your contribution

WHAT YOU ACTUALLY DO

1. Re-derive the key equations; check dimensions and assumptions.
2. Reconcile notation between paper and code (they often disagree).
3. Clone the repo. Run it **as published**, then on **your** data.
4. Record every discrepancy: undocumented preprocessing, different hyperparameters in code vs paper, a metric implemented non-standardly.
5. Read supplementary material and appendices — the honest details live there.
6. Read **OpenReview** reviews/rebuttals if available.
7. Email the authors with one precise question. Most reply.

WHAT A DEEP READ PRODUCES

- A **trustworthy baseline number** you can defend to a reviewer
- A list of **hidden assumptions** = candidate research gaps
- Sometimes: a **reproducibility finding**, which is itself publishable

Real pattern, worth stating plainly: the reason reported numbers often cannot be reproduced is rarely fraud. It is usually undocumented preprocessing, different splits, unequal tuning budgets, or a non-standard metric implementation. Documenting *which* one it was is a contribution.

Report reproduction failures **about the method, not the authors**: "we could not reproduce the reported 0.91 AUC using the released code with the described protocol; we obtained 0.86 ± 0.01 over 5 seeds," plus your exact configuration.



The Paper Extraction Template HANDOUT

Fill **one per paper** in Pass 2. These 14 fields become one row of your literature matrix (S6) and the raw material of your related-work section (Day 2, S13).

#	Field	What to capture
1	Citation key + venue + year	Author2025Domain ; venue; indexed?
2	Research problem	The unknown they address, in <i>your</i> words
3	Motivation	Why it matters (application or theory)
4	Stated objective	Their explicit aim/RQ
5	Claimed gap	What they say prior work lacked
6	Dataset(s)	Names, size, classes, split protocol, public?
7	Preprocessing	Resizing, normalisation, augmentation, balancing

#	Field	What to capture
8	Proposed method	Architecture/algorithm; the <i>new</i> component
9	Baselines	What they compared against; tuned equally?
10	Metrics	Which, and is the choice appropriate?
11	Experimental setup	Seeds, epochs, optimiser, LR, hardware, CV
12	Key results	2–3 numbers with the comparison point
13	Limitations	Theirs (quote verbatim) + yours (what they missed)
14	Future work + gap for me	Their suggestions + your one-line opportunity

Discipline rule: field 13 must contain **both** their admitted limitation *and* your independent critique. A template with only the authors' self-assessment reproduces their framing — that is summarising, not reviewing.

Worked Example — Extraction of One Paper

Paper P7 (*illustrative composite constructed for this workshop — not a real citation*): "Site-Adversarial Training for Cross-Hospital Chest Radiograph Classification", *hypothetical journal*, 2024.

Field	Extracted content
Problem	CXR classifiers lose accuracy on unseen hospitals; degradation unquantified under site-wise splits
Motivation	Deployment across hospitals with different scanners/protocols
Objective	Reduce cross-site AUC drop without harming in-domain AUC
Claimed gap	Prior DG work evaluated on random splits, which leak site identity
Datasets	CheXpert (train), NIH ChestX-ray14 (external), MIMIC-CXR (external); site-wise splits
Preprocessing	224×224, histogram equalisation, random crop/flip; no lung segmentation
Method	DenseNet-121 + gradient-reversal site-discriminator branch (λ ramped 0→1)
Baselines	ERM DenseNet-121, IRM, CORAL, Mixup
Metrics	Macro AUC, worst-site AUC, ECE
Setup	3 seeds, Adam 1e-4, 30 epochs, 1×A100; code released
Results	Worst-site AUC 0.78 → 0.83; in-domain AUC 0.89 → 0.88; ECE improved 0.09 → 0.05

FIELD 13 — LIMITATIONS

Theirs (verbatim-style): "restricted to frontal adult radiographs; site labels assumed known at training time."

Mine (independent critique):

- Only **3 seeds**; the 0.05 worst-site gain has **no significance test** or CI
- **Site labels required** — unavailable in most real deployments
- No **subgroup** analysis (age, sex, device); worst-site may hide worst-group
- No cost report (training time/params)
- ECE improvement not analysed — is it the adversary or just the extra regularisation? **No ablation on λ**

FIELD 14 — FUTURE WORK → MY OPPORTUNITY

Theirs: extend to lateral views and paediatric data.

My one-liner: *Site labels are the binding constraint. Can unsupervised site clustering recover most of the worst-site gain without site labels? And does the reported gain survive 10 seeds with a paired test?*



Section 4 — Wrap-Up

COMMON MISTAKES

- Reading linearly from page 1
- Highlighting instead of extracting into fields
- Believing the abstract's number without checking the table
- Recording only the authors' limitations, never your own
- Never checking whether the paper was refuted by later work
- Letting an AI summary replace the read for a **core** paper
- Not recording the citation key at extraction time → hours lost later

TOOLS **TOOLS**

Zotero + Zotfile/tags · Semantic Scholar (TLDR, influential citations) · Connected Papers · OpenReview · Papers with Code · SciSpace / Elicit / NotebookLM (*assistive, verify everything*) · Excel/Sheets for the triage log

YOUR READING DASHBOARD TARGET

Stage	Count	Depth
Retrieved	80–300	metadata only
Triaged (Pass 1)	all	5–10 min each
Core set (Pass 2)	10–20	template filled
Deep read (Pass 3)	3–6	code run

Takeaway: Efficient reading is not faster reading — it is **selective depth**. Triage everything, comprehend the core, reconstruct the few papers your contribution stands on.



Section 5

Literature Review

Synthesis, not summary: writing that argues rather than lists

What a Literature Review Is — and Is Not

A literature review is an argument about a body of work that (a) organises it into a structure of your making, (b) evaluates its evidence, and (c) demonstrates that a specific gap exists.

WHY IT IS REQUIRED

Function	Consequence if missing
Proves the gap is real	"This has been done before" → reject
Positions your contribution	Reviewer cannot see novelty
Justifies your method choices	"Why this baseline/metric?"
Identifies your baselines	Missing comparison → major revision
Shows command of the field	Loss of credibility; viva failure
Chooses your reviewers	They <i>will</i> be among the authors you cite

SUMMARY VS REVIEW

	Summary (paper-by-paper)	Review (synthesis)
Unit of a paragraph	One paper	One idea, method family, or finding
Order	Whatever order you read them	Your taxonomy
Verbs	"proposed", "used", "achieved"	"converge on", "diverge", "in contrast", "remains untested"
Comparison	None	Explicit, on shared dimensions
Ends with	The last paper	The gap
Citations per sentence	1	Often 2–5 grouped

The tell-tale sign of a summary: every paragraph starts with an author's name.

Five Ways to Organise a Review

Structure	Organising principle	Use when	Risk
Chronological	Time: early → recent	The field has clear paradigm shifts (HMM → RNN → Transformer)	Becomes a timeline with no argument
Thematic	Sub-problems / themes	Several distinct facets (data, method, evaluation, deployment)	Themes overlap; papers repeat
Methodological	Technique families	You will argue that a <i>family</i> has a shared weakness	Ignores task differences
Comparative	Dimensions in a matrix	You have quantitative evidence across papers	Reads like a spreadsheet if not interpreted
Critical / argumentative	Your thesis about the field	You have a strong, defensible position	Needs deep mastery; can look biased

WHAT STRONG REVIEWS ACTUALLY DO

Hybrid: a **thematic** top level, **methodological** groups inside each theme, a **comparative table** per theme, and a **critical** synthesis paragraph closing each theme — with the last paragraph of the section stating the gap.

SKELETON YOU CAN COPY (RELATED WORK, ~4 SUBSECTIONS)

- 2.1 Problem formulations & datasets
- 2.2 Method family A (+ table, + critique ¶)
- 2.3 Method family B (+ table, + critique ¶)
- 2.4 Evaluation practices in this area
- 2.5 Synthesis: what is settled, what is contested, what is untested ► THE GAP



Weak vs Strong: The Same Five Papers

✗ WEAK (SUMMARY CHAIN — THE MOST COMMON RELATED-WORK FAILURE)

"Sharma et al. [3] used a CNN for chest X-ray classification and achieved 88% accuracy. Wang et al. [4] used DenseNet-121 and achieved 90%. Li et al. [5] proposed a transformer-based model with 91% accuracy. Kumar et al. [6] used CORAL for domain adaptation and reported improvement. Ahmed et al. [7] used IRM and achieved better generalisation."

Diagnosis: five facts, zero relationships. No shared dimension (different datasets and splits, so the accuracies are not comparable). No evaluation of quality. No gap. The reader cannot tell what is *known*.

✓ STRONG (SYNTHESIS WITH A CRITICAL EDGE)

"Work on cross-hospital CXR classification splits along two lines. The first improves **representation capacity** — DenseNet variants [4] and vision transformers [5] — and reports 88–91% in-domain accuracy, but *all three studies evaluate on random splits in which images from the same institution appear in both training and test sets*, so their numbers describe in-distribution performance rather than transfer [3]–[5]. The second line explicitly targets **distribution shift** using invariance objectives such as CORAL [6] and IRM [7]; these do evaluate across institutions and report 3–6 point gains in external AUC, but both assume **site labels are available at training time** and neither reports variance over more than three seeds, so it is unclear whether the reported gains exceed run-to-run noise. Across all five studies, **worst-site** performance and **calibration** go unreported, although these are the quantities that determine clinical usability. The literature therefore establishes *that* shift matters, but not *how large* the degradation is under leakage-free protocols, nor whether invariance methods help when site labels are unavailable."



The Synthesis Language Toolkit

AGREEMENT / CONVERGENCE

- "Consistently across [3]–[7], ..."
- "There is broad agreement that ..., though the effect size varies from X to Y."
- "Both lines of work converge on ..."

CONTRAST / CONTRADICTION

- "In contrast to [4], who report ..., [8] find the opposite when ..."
- "This discrepancy is plausibly attributable to differing split protocols rather than to the methods themselves."
- "Whereas early work assumed ..., more recent evidence suggests ..."

EVALUATION OF QUALITY

- "..., although the comparison uses an untuned baseline."
- "...; the reported gain (0.4 points) is smaller than the seed-to-seed variance reported in [9]."
- "..., on a single dataset, which limits generality."

ESTABLISHING ABSENCE (GAP MOVES)

- "To the best of our knowledge, no study has ..." (*only after a documented search*)
- "X remains unquantified under [condition]."
- "Existing evaluations do not report [quantity], although it determines [consequence]."
- "This assumption ([assumption]) is unlikely to hold in [setting], yet is untested."

Never write "no work exists" casually. Reviewers *will* find a counterexample. Safer, and stronger: *"we found no study that evaluates X under Y; the closest is [12], which does Z but not Y."* This shows you looked and it survives contact with a specialist.

CITATION HYGIENE

- Group citations by claim: [3]–[5] not one per sentence
- Cite the **origin** of a method, not only a recent user
- Never cite a paper you have not opened



Section 6

The Literature Matrix

The single artefact that turns 15 PDFs into a research gap

Matrix Columns

One row per paper. Build it in a spreadsheet — **not** in a Word document.

Col	Column name	Fill with	Why it earns its place
A	Paper (citation key)	Wang2023DenseCXR	Links matrix → Zotero → BibTeX
B	Year	2023	Chronology; recency of the frontier
C	Venue + indexing	IEEE TMI, SCIE Q1	Weights evidence quality; hints at your target journal
D	Research problem	Cross-site degradation	Groups papers into themes
E	Dataset(s) + split protocol	CheXpert; random split	The most diagnostic column — reveals leakage & incomparability
F	Method	DenseNet-121 + CORAL	Builds your method taxonomy
G	Baselines	ERM only	Reveals weak-comparison papers
H	Metrics	Accuracy, AUC	Reveals evaluation gaps
I	Key results	0.90 AUC in-domain	Comparable <i>only</i> within same dataset+split
J	Strengths	Public code; 3 datasets	Fair credit; tells you what to emulate
K	Limitations (theirs + yours)	Random split; 1 seed	Raw material for your gap
L	Research gap it leaves	Leakage-free re-evaluation missing	Aggregated → your gap
M	Future work (verbatim)	"extend to lateral views"	Free, author-endorsed ideas
N	Reproducibility	Code ✓ / weights ✗ / seeds 1	Feasibility of using it as a baseline
O	Relevance to me	Baseline / context / discard	Sorting and prioritisation



Filled Matrix Extract (Illustrative)

Paper	Yr	Dataset + split	Method	Baselines	Metrics	Key result	Limitation (mine)	Gap it leaves
P1	2022	CheXpert, random	ResNet-50	none	Acc, AUC	0.89 AUC	Leakage across patients; 1 seed	In-domain only; no transfer claim possible
P2	2022	NIH14, random	DenseNet-121	ResNet-50	AUC	0.90 AUC	Baseline untuned; no CI	Fair-tuning comparison missing
P3	2023	CheXpert→NIH14, site-wise	ViT-B/16	DenseNet	AUC	0.84 external	1 external set; no worst-site	Multi-site breadth
P4	2023	MIMIC-CXR, random	CNN+attention	2 CNNs	Acc, F1	0.91 Acc	Accuracy on imbalanced multi-label	Metric inappropriate → PR-AUC needed
P5	2023	CheXpert→NIH14	CORAL	ERM	AUC	+0.03 external	Needs site labels; 3 seeds	Label-free adaptation
P6	2024	3 sets, site-wise	IRM	ERM, CORAL	AUC, ECE	+0.04 worst-site	No significance test	Statistical validation
P7	2024	3 sets, site-wise	Site-adversarial	ERM, IRM, CORAL, Mixup	AUC, worst-site, ECE	0.78→0.83 worst-site	Site labels; 3 seeds; no λ ablation	Label-free + ablation + stats
P8	2024	Private, undisclosed	Ensemble	1 CNN	Acc	0.95 Acc	Not reproducible; no access	— (discard as baseline)
P9	2025	CheXpert, random	Self-supervised pretrain	ERM	AUC	+0.02	No external validation	SSL under shift untested
P10	2025	2 sets, site-wise	Test-time adaptation	ERM, CORAL	AUC	+0.03	Needs target batch; latency unreported	Deployment cost of TTA

Read the columns, not the rows. Column patterns are the finding: 6/10 used random splits · 4/10 compared against a single baseline · 0/10 ran a significance test · 2/10 reported calibration · 0/10 worked without site labels.

Building the Matrix — Tools DEMO

Tool	Purpose	How to use	Limitation / verification
Excel / Google Sheets	The matrix itself	Row 1 = headers, freeze ; one row per paper; Data → Create a filter ; conditional formatting to colour split protocol; COUNTIF for column tallies; Sheets = collaborative + version history	Manual entry; keep one authoritative copy to avoid fork chaos
Zotero (free, open)	Reference library + PDFs + notes	Import the RIS/BibTeX from S3 → <i>Add Item by Identifier</i> for DOIs → colour tags (core/context/discard) → child notes using the extraction template → <i>Export Collection</i> → <i>CSV</i> to seed the matrix	Metadata from publishers is often wrong (missing pages, wrong capitalisation) — fix every field manually
Mendeley	Same role, Elsevier ecosystem	Web importer; Mendeley Cite for Word; annotate PDFs	Fewer plugins than Zotero; account-bound
EndNote	Institutional standard in many labs	Groups; Cite While You Write	Paid; heavier
Semantic Scholar / Elicit / SciSpace / NotebookLM	AI-assisted extraction	Upload your PDF set; ask for dataset/metric/baseline columns; export a draft table	VERIFY Every extracted number, dataset name and claim must be checked against the PDF. AI tools mis-attribute numbers between tables and hallucinate absent fields. Treat output as a first draft of a data-entry task , never as evidence
Connected Papers / Litmaps /	Find what your matrix is missing	Seed with 3–5 core papers; look for prominent nodes you have not read	Coverage depends on the underlying graph

Division of labour rule for the whole workshop: AI may help you **find, sort, translate, and pre-fill**. You must **read, verify, compare, judge, and claim**.



Section 6 — Activity and Wrap-Up

ACTIVITY 6.1 — 45 MIN (CORE ACTIVITY OF DAY 1)

Build your literature matrix

1. From your S3 shortlist, load **10–15 papers** into Zotero. **5 min**
2. Pass-1 triage all of them; discard the irrelevant, replace them to keep ≥ 10 . **15 min**
3. Fill the matrix: columns A–O. Use AI assistance for E–I if you like, then **verify each cell against the PDF**. **20 min**
4. Compute **column tallies** for split protocol, number of baselines, metrics, significance testing, code availability. **5 min**
5. Write the five tallies as five sentences. (*These become your gap paragraph in Section 7.*)

Deliverable: one spreadsheet + five tally sentences.

COMMON MISTAKES

- Building the matrix in Word (cannot sort, filter or tally)
- One column called "Notes" instead of 15 specific columns
- Averaging results across incomparable datasets/splits
- Recording only the authors' limitations
- Trusting AI-extracted numbers without opening the PDF
- Not recording the split protocol — the highest-value cell
- Stopping at 5 papers (patterns are invisible below ~10)

Takeaway: The matrix converts reading into *data*. Gaps are then found by **counting columns**, not by hoping for inspiration.



Section 7

Finding the Research Gap

The section that decides whether your paper is publishable

What a Research Gap Is

A research gap is a specific, evidenced absence in the literature whose resolution would change what the field knows or can do.

A GAP STATEMENT HAS FOUR OBLIGATORY PARTS

Part	Example
What is absent	No leakage-free, multi-site quantification of CXR classifier degradation
Evidence of absence	11 of 15 studies use random splits; 0 of 15 report worst-site performance
Why it matters	Reported AUCs overstate deployable performance; hospitals cannot judge risk
What resolving it enables	A trustworthy degradation estimate + a label-free mitigation

THESE ARE NOT GAPS

Non-gap	Why
"Nobody has applied model X to dataset Y"	Absence of an activity $\neq$ absence of knowledge. Ask <i>why it would be informative</i>
"Accuracy can be improved"	Always true, everywhere, forever
"The topic is new/trending"	Novelty of hype $\neq$ gap
"Existing methods are complex"	Only a gap if you show complexity <i>costs</i> something measurable
"No work exists in my country/language"	Becomes a gap only with a mechanism: <i>why</i> would results differ here?
"No one has combined A and B"	Combination is not a contribution unless you show <i>why</i> the combination should behave differently



Gap Taxonomy (1 of 2)

#	Gap type	Definition	AI/ML example	Typical evidence you need
1	Knowledge gap	A phenomenon is unexplained or unmeasured	Why does self-supervised pretraining help under shift but not under label noise?	Prior work reports the effect but not the cause
2	Methodological gap	Existing methods rest on an assumption that fails in practice	All DG methods for CXR require site labels ; deployments do not have them	Assumption stated in each paper's setup
3	Dataset / resource gap	No suitable data exists: language, modality, population, annotation	No code-mixed Hindi-English paraphrase benchmark with obfuscation levels	Search of dataset registries returns nothing
4	Performance gap	Best known result is insufficient for the use case	Worst-site AUC 0.83 vs the ≥ 0.90 needed for triage support	Requirement from the application domain
5	Application / translation gap	Method works in the lab, unstudied under deployment constraints	Test-time adaptation needs target batches; latency/memory unreported on edge devices	Absence of cost reporting in prior work
6	Population / domain gap	Evidence exists for one group/domain and is untested for another, <i>with reason to expect difference</i>	Models trained on adult frontal CXR untested on paediatric; anatomy differs	Domain argument + absence of studies



Gap Taxonomy (2 of 2)

#	Gap type	Definition	AI/ML example	Typical evidence you need
7	Evaluation gap	Metrics or protocols do not measure what matters	Multi-label imbalanced CXR evaluated by accuracy; PR-AUC and calibration unreported	Column tally of metrics across papers
8	Scalability gap	Works at small scale; behaviour at target scale unknown	Graph method validated to 10^4 nodes; production graphs are 10^8	Complexity analysis + absence of large-scale results
9	Generalisation gap	Results do not transfer across distributions	Random-split AUC 0.90 → site-wise 0.78	Direct re-evaluation
10	Reproducibility gap	Results cannot be independently obtained	6/15 papers release no code; reported numbers not recoverable	Your own reproduction attempt, documented
11	Computational-efficiency gap	Accuracy achieved at unreported/unacceptable cost	+0.02 AUC for 4× parameters and 3× latency — never reported	Cost measurements absent in prior work
12	Explainability / trust gap	Decisions unexplained, or explanations unvalidated	Grad-CAM used but never checked against expert annotation	Absence of explanation-validation studies

Strategic note for early-career researchers: gaps 7, 9, 10, 11 are usually the cheapest to address rigorously (they need careful experiments, not new theory or big compute), and they are hard for reviewers to dismiss because the evidence is arithmetic.

Where Gaps Hide: Nine Signals

Signal in the literature	How to detect it	Gap type it usually yields
Stated limitations	Read every Limitations/Threats section; tally recurring ones	2, 5, 6, 12
Future work	Collect verbatim; a suggestion repeated by 3+ papers is a field priority	any
Contradictory results	Two papers, same task, opposite conclusions → find the confound (data? split? tuning?)	1, 7, 9
Missing datasets	Search dataset registries; note languages/populations/modalities absent	3, 6
Poor evaluation	Tally metrics column: accuracy on imbalanced data, no CI, no significance test	7
Weak baselines	Tally baselines column: "compared to ERM only", or to a 5-year-old method	4, 7
Missing comparisons	Method families that have never been compared <i>under one protocol</i>	7, 9
Small samples / few seeds	n < 100 subjects; 1–3 seeds; single split	7, 10
Domain/scope limitations	"We consider only English/adults/frontal views/synthetic data"	6, 8, 9

Contradictions are the richest signal and the most under-used. When P4 reports that attention helps and P9 reports it does not, the *explanation* of the discrepancy is a knowledge contribution — and you can often produce it with a controlled re-run rather than a new method.

From 10 Papers to a Gap (1/2) — Counting

STEP 1 · 15 papers in the matrix (2022–2026)

STEP 2 · COLUMN TALLIES

- 11/15 random splits (patient/site leakage)
- 12/15 report only accuracy or AUC
- 0/15 report worst-site performance
- 2/15 report calibration (ECE)
- 0/15 report any significance test
- 9/15 compare against ≤ 1 baseline
- 6/15 release no code
- 15/15 DG methods require site labels

STEP 3 · COMMON LIMITATIONS

L1 Evaluation leaks site/patient identity

(11/15) → in-distribution numbers only

L2 Reported gains within seed noise

(0/15 significance tests, ≤ 3 seeds)

L3 Site labels assumed available

(15/15) → untestable in deployment

L4 Clinically relevant quantities

unreported (worst-site 0/15, ECE 2/15)

THE MOVE

Each **L** is a *field-level* limitation, not one author's excuse — because it is a **count across papers**, taken from one matrix column.

Do this in a spreadsheet, not in your head. The tallies are your evidence; you will cite them in your Introduction (¶4) and your reviewers will check them.

Rule of thumb: a weakness admitted or exhibited by $\geq 40\%$ of your set is a gap candidate. A weakness in one paper is a critique of that paper.



From 10 Papers to a Gap (2/2) — Converting

STEP 4 · UNRESOLVED PROBLEM

The true cross-hospital degradation of CXR classifiers is unknown, and no mitigation has been shown to work WITHOUT site labels.

STEP 5 · RESEARCH GAP (evidenced)

No study quantifies degradation under leakage-free site-wise protocols across multiple institutions with statistical validation, and no label-free mitigation has been evaluated on worst-site AUC and calibration.

↑ from L1, L2 ↑ from L3, L4

STEP 6 · PROPOSED CONTRIBUTION

C1 Leakage-free multi-site re-evaluation of 5 published models (3 datasets, 10 seeds, paired tests, CIs) ← L1, L2

C2 A label-free site-clustering adaptation method; ablation isolates the mechanism ← L3

C3 Public benchmark + code + splits ← L1, L4

WHY THIS CHAIN CONVINCES A REVIEWER

- Every step is **countable**, from a documented search
- The gap is **not** "accuracy can be improved"
- Contributions map **1:1** to the limitations they answer
- Two of three contributions need **no new theory** and modest compute
- A **negative** result is still publishable: if degradation turns out to be small, that is important news

Notice: contributions C1 and C3 were *created by the tallies*. **The gap analysis designed the study** — the method was chosen last, not first.



Weak vs Strong Gap Statements

✗ WEAK

1. "Many researchers have worked on CXR classification, but there is still room for improvement."
2. "Deep learning has not been applied to our hospital's dataset."
3. "Existing methods have low accuracy."
4. "No one has combined transformers with domain adaptation for CXR."
5. "Research on explainability in medical imaging is limited."
6. "Prior work is old; we use a newer model."

Why each fails

1. Vague; no absence identified; unfalsifiable
2. Local activity, not knowledge; no mechanism
3. Unquantified; low relative to what threshold?
4. Untried combination without a reason to expect different behaviour
5. "Limited" is not evidence; how limited, in what respect?
6. Recency is not a contribution

✓ STRONG

1. "Across 15 studies (2022–2026), 11 evaluate CXR classifiers on random splits in which images from the same institution occur in both partitions; consequently the reported AUCs (0.88–0.91) cannot be interpreted as cross-hospital performance, and the magnitude of degradation under leakage-free protocols remains unquantified."
2. "All eight domain-generalisation methods we identified require site labels during training [5]–[7], [11]; in deployment, provenance metadata is typically stripped for privacy, so it is unknown whether comparable gains are attainable without site supervision."
3. "None of the 15 studies reports worst-site AUC or calibration error, although both determine clinical usability; performance is therefore reported for the average institution and not for the institution most at risk."

THE TEMPLATE

"Across *[N]* studies identified by *[documented search]*, *[pattern]* holds. Consequently, *[specific quantity/mechanism]* remains *[unquantified / untested / unexplained]*, even though *[why it matters]*. This study addresses that by *[action]*."



Section 7 — Activity and Wrap-Up

ACTIVITY 7.1 — 30 MIN

Derive your gap (worksheet: [handouts/research-gap-worksheet.md](#))

1. Tally 5 columns of your matrix. **5 min**
2. Write 3–5 **common limitations** (L1...L5), each with the count and 2–3 supporting citations. **8 min**
3. State the **unresolved problem** in one sentence. **3 min**
4. Write the **gap** using the template, including numbers. **7 min**
5. Classify it against the **12 gap types** (it may be 2–3 types). **2 min**
6. List **2–3 contributions** mapping 1:1 to the gap. **5 min**
7. **Stress test with a partner:** the partner must attempt to break it with — *"has this been done?", "how do you know?", "why does it matter?", "can you do it with your resources?"*

COMMON MISTAKES

- "Room for improvement" as a gap
- A gap with no numbers
- Claiming "no work exists" without a documented search
- A gap that requires resources you do not have
- Gap and contributions that do not correspond
- Finding the gap **after** choosing the method
- One gap type only, when the honest answer is "evaluation + generalisation"
- Never stress-testing it with a critical reader

Takeaway: A research gap is not discovered by inspiration. It is **computed** from a documented literature matrix, stated with numbers, and stress-tested by a hostile reader before it reaches a reviewer.



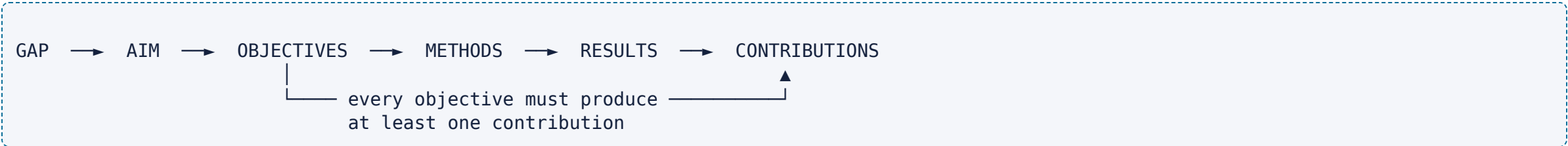
Section 8

Research Objectives and Contributions

Turning a gap into a work plan a reviewer can audit

Aim → Objective → Method → Contribution

Element	Count	Function	Grammar	Example
Aim	Exactly 1	The overall purpose; broad but bounded	"To + purpose"	"To establish how much chest-radiograph classifiers degrade across hospitals and whether label-free adaptation mitigates it."
Objective	3-5	Concrete, completable steps that jointly achieve the aim	"To + measurable verb + object + condition"	"O2: To quantify the AUC and calibration degradation of five published models under site-wise splits on three datasets over 10 seeds."
Method	per objective	How the objective will be executed	Procedure	"Re-train each model with identical augmentation, tuning budget and seeds; evaluate with macro/worst-site AUC, ECE; compare with paired Wilcoxon tests + bootstrap CIs."
Contribution	2-4	What the field gains once objectives are met	Noun phrase, calibrated	"C1: The first leakage-free multi-site degradation benchmark for CXR classification, with statistical validation."



Traceability audit (reviewers do this): each objective → a method → a results subsection → a contribution bullet. An objective with no results subsection reads as an abandoned promise; a results subsection with no objective reads as an afterthought.

Gap → Research Questions → Objectives

THE CONVERSION IS MECHANICAL

Gap element	Becomes
"...magnitude of degradation is unquantified"	RQ1 How large is it?
"...requires site labels"	RQ2 Can it be done without them?
"...worst-site and calibration unreported"	RQ3 What happens to those quantities?
"...gains may be within noise"	RQ4 Are differences statistically significant?

RQ QUALITY RULES

- **One** relationship per RQ (two variables)
- Answerable by an experiment you can actually run
- Not answerable by yes/no alone → prefer *how much, under what conditions, why*
- Ordered: descriptive (RQ1) → comparative (RQ2) → explanatory (RQ3)

FROM RQ TO OBJECTIVE — WORKED

RQ1	How much does classification performance degrade from random to site-wise evaluation?
O1	To re-evaluate five published architectures on three public datasets under both random and site-wise protocols, over 10 seeds, reporting macro AUC, worst-site AUC and ECE with 95% bootstrap CIs.
RQ2	Can degradation be mitigated without site labels?
O2	To develop and evaluate a label-free adaptation method based on unsupervised site clustering, compared against ERM, CORAL, IRM and site-adversarial training under an identical tuning budget.
RQ3	Which component drives the mitigation?
O3	To conduct an ablation over the clustering granularity, the invariance weight λ , and the augmentation set, isolating each component's effect.



Hypotheses and Contribution Calibration

Hypotheses (state them even if you never print them)

Form	Example
H0 (null)	Site-wise and random-split AUC do not differ (paired, $\alpha=0.05$).
H1 (alternative, directional)	Site-wise AUC is lower than random-split AUC.
H2	Label-free clustering adaptation improves worst-site AUC by ≥ 0.03 over ERM.

RULES

- Must be **falsifiable** and **pre-specified** (before you look at test results)
- Name the **test** and **α** in advance: paired *t*-test / Wilcoxon signed-rank / McNemar (same test set, two classifiers)
- Include an **effect size**, not only significance: 0.001 gain can be significant and useless
- Correct for **multiple comparisons** (Holm/Bonferroni) when testing many models $\times$ datasets

Choosing the test *after* seeing the results, or reporting only the comparisons that reached significance, is **p-hacking** — a form of misconduct (S21).

Calibrating contribution claims

✗ Over-claim	✓ Calibrated
"We propose a novel state-of-the-art architecture"	"We propose a label-free adaptation module that improves worst-site AUC by 0.04 ± 0.01 over the strongest baseline on three datasets"
"Our method solves domain shift"	"Our method recovers ~60% of the degradation observed under site-wise evaluation"
"First work in this area"	"To our knowledge, the first leakage-free multi-site evaluation of these five architectures"
"Outperforms all existing methods"	"Outperforms four representative baselines under an identical tuning budget; we did not evaluate methods requiring paired multi-site supervision"

Calibrated claims are stronger, not weaker. A number with a CI and a scope limit is unassailable; "state of the art" invites a reviewer to find the counterexample — and they will.



Weak vs Strong Objectives

✗ Weak objective	Diagnosis	✓ Strong objective
"To study deep learning for medical imaging"	No boundary, no completion criterion	"To quantify the change in macro and worst-site AUC of five published CXR architectures when random splits are replaced by site-wise splits, across three public datasets and 10 seeds"
"To implement a CNN model"	Activity, not knowledge; done in a day	"To determine whether label-free site clustering recovers $\geq 50\%$ of the site-wise degradation achieved by site-supervised adversarial training"
"To improve accuracy"	Unquantified, no comparison point	"To reduce worst-site AUC degradation by ≥ 0.03 relative to ERM at equal parameter count and inference latency"
"To analyse various algorithms"	"Various" = unspecified	"To compare ERM, CORAL, IRM and site-adversarial training under an identical 50-trial tuning budget on identical splits"
"To develop a novel framework"	Unfalsifiable; "framework" hides the claim	"To develop an adaptation module requiring no site metadata and evaluate it on worst-site AUC, ECE and inference latency"
"To achieve 99% accuracy"	Fixates on a number; unachievable/meaningless target	"To characterise the accuracy-latency trade-off across four model scales on a Jetson-class device"

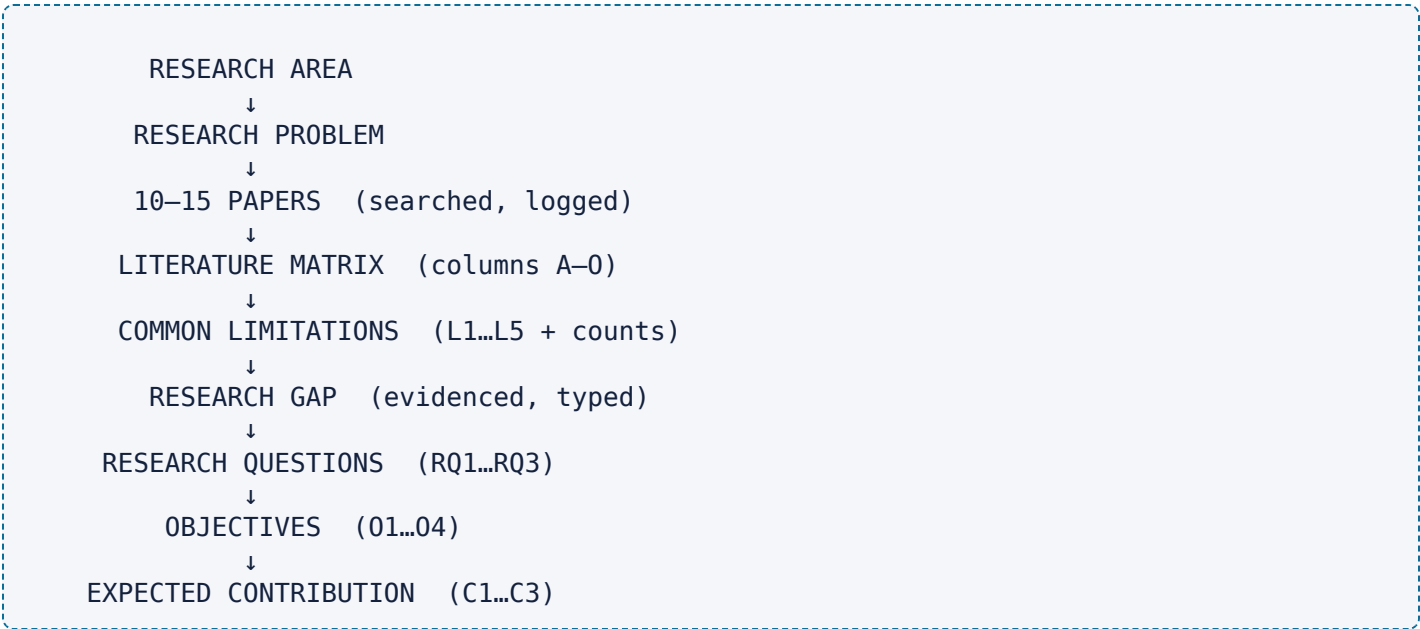
Self-check per objective: Does it name a **measurable outcome**? A **comparison**? A **condition/scope**? Could an examiner mark it **done or not done**? Does it map to a **contribution**?

Day 1 Hands-On Activity

From Area to Expected Contribution

90 minutes of work, then presentations

Day 1 Capstone — Brief



TIMING (90 MIN)

Min	Task
0–10	Finalise area + problem statement
10–25	Finalise the paper set (≥10, logged strings)
25–55	Complete the matrix; compute tallies
55–70	Limitations → unresolved problem → gap
70–85	RQs, objectives, contributions, traceability table
85–90	Prepare a 3-slide / 3-minute presentation

- SUBMISSION FORMAT (ONE FILE, `LASTNAME_DAY1.MD/.DOCX` + THE MATRIX SPREADSHEET)**
1. Area, sub-area, problem (3 sentences)
 2. Search log: databases, strings, dates, counts, screening funnel
 3. Matrix: ≥10 rows × 15 columns
 4. Five tally sentences
 5. Common limitations L1–L5 with counts and citations
 6. Gap statement (template) + gap type(s)
 7. RQ1–RQ3
 8. Aim + objectives O1–O4
 9. Hypotheses with named tests and α
 10. Contributions C1–C3
 11. Traceability table
 12. Feasibility line: datasets + licences, baselines + repos, GPU-hours, killer risk

Rule: every number in your submission must be traceable to a matrix cell, and every citation must be one you have actually opened.



Presentations and Rubric

3-MINUTE PRESENTATION FORMAT

Slide	Content	Time
1	Problem + why it matters	45 s
2	Matrix tallies → gap (with numbers)	90 s
3	RQs, objectives, contributions	45 s

AUDIENCE JOB (MANDATORY)

Each presentation gets **two questions**, drawn from:

- "Has this already been done? Which paper is closest?"
- "How do you know the gap is real — which count supports it?"
- "Who benefits if you succeed?"
- "What is your riskiest assumption?"
- "Can you run this with your compute?"
- "What would a negative result look like, and would it be publishable?"

RUBRIC (20 POINTS)

Criterion	Pts
Search is documented and reproducible (strings, dates, counts)	3
Matrix ≥10 rows, columns complete, split protocol recorded	4
Limitations supported by counts , not impressions	3
Gap is specific, evidenced, and correctly typed	4
Objectives are measurable and mark-able done/not-done	3
Contributions map 1:1 to the gap and are calibrated	2
Feasibility stated honestly (data, licence, compute, risk)	1

DAY 1 TAKEAWAY

You now hold the two artefacts that decide a paper's fate long before writing begins: a **literature matrix** and an **evidenced research gap**. Tomorrow we turn them into a manuscript.

